04 The Drift Room

The Wright–Fisher model

CORE PATH · BSC	EXTENSION · MSC	BEFORE THIS ROOM
about 35 min	**+ 34 min**	The Fate Room (03) is the natural lead-in: it followed one new mutation, this room starts from any frequency.
steps 1–4	steps 5–7	

OPEN THE ROOM

`https://evolinus.github.io/zooe/teaching/evolutionary-laboratories/#drifting`

A population of N carries two alleles, A1 and A2, and neither is better than the other. Every generation is drawn at random from the one before. This guide takes you from a single run to the one thing drift is reliable about: how fast it removes variation.

LEARNING OBJECTIVES

After the core path you should be able to

- Explain why an allele frequency changes in a finite population even when no allele has an advantage.
- Predict how population size N changes the size and speed of those changes, and test the prediction.
- State the neutral probability of fixation, $u(p) = p$, and check it against a batch of runs.
- Explain why fixation and loss are permanent when there is no mutation.

The extension adds

- Use $\mathrm{Var}(\Delta p) = p(1-p)/M$ and $H_t = H_0(1 - 1/M)^t$ to predict a run before making it.
- Separate what drift does unpredictably (which allele wins, and when) from what it does predictably (how fast heterozygosity falls).
- Explain how recurrent mutation turns decay into a mutation–drift balance.

SUGGESTED ROUTES

- **45-minute practical (BSc):** steps 1–4, then questions 1–3 as a group discussion.
- **Self-study:** the core path, then the *Checking your reasoning* page. Allow 40 minutes.
- **MSc seminar:** the core path as preparation; class time on steps 5–7 and questions 4–6.
- **15-minute demonstration:** step 2 projected, with the class voting on the prediction before each run.

How to work through it. Every step asks you to *predict* before you press anything, and to *explain* afterwards. Write the prediction down and commit to it: a wrong prediction you can account for teaches more than a right one you guessed. Runs are random, so where a step says so, run it more than once before you decide what happened.

1 Watch one generation being made

8 min

SET UP

The defaults: N = 50, initial p = 0.50, Haploid, mutation rate μ = 0.

PREDICT FIRST

The wheel will draw 50 alleles from a population that is exactly half A1. Will the next generation be exactly half A1 again? If not, how far from 0.50 could it plausibly land?

RUN

Press **Run Simulation**. The first two generations are drawn on the Sampling Wheel one individual at a time; press **Pause after gen** if you want to stop and look.

LOOK AT

the p = *freq. A1* readout after generation 1, and the Binomial Distribution panel beneath the chart.

EXPLAIN

Neither allele is better. Where, then, did the change in p come from? Use the binomial panel to say what range of outcomes was likely.

2 Small against large

10 min

SET UP

p = 0.50, Haploid. First N = 10, then N = 200. Leave Generations G alone: it follows N until you move it.

PREDICT FIRST

Sketch the trajectory you expect at N = 10 and at N = 200. Which reaches 0 or 1 first, and after roughly how many generations?

RUN

Run at N = 10, then at N = 200, changing nothing else. Run each one twice more.

LOOK AT

the *Allele Frequency (p) over Generations* chart and the *Fixation & loss* readout.

EXPLAIN

Why does the small population lurch and the large one barely move? Which allele ended up fixed — and could you have predicted *which* from the settings?

3 Ten populations at once

10 min

SET UP

$N = 50$, $p = 0.50$, Haploid.

PREDICT FIRST

Of ten independent populations starting at $p = 0.5$, how many will end with A1 fixed? Now the same question for $p = 0.2$.

RUN

Press **Run 10 Simulations** at $p = 0.5$ and count how many runs fixed A1, lost it, or were still polymorphic at the end. Repeat at $p = 0.2$.

LOOK AT

where the ten lines end, and how many of them have finished.

EXPLAIN

Compare your counts with $u(p) = p$. Why does one batch of ten match it only roughly? If you are in a class, pool everyone's counts and compare again.

4 Can a lost allele come back?

7 min

SET UP

$N = 20$, $p = 0.50$, $\mu = 0$.

PREDICT FIRST

Once a run reaches $p = 0$, what will p do in the generations that follow?

RUN

Run until A1 is fixed or lost, and let the run carry on. Then drag the History Scrubber back to the generation where it happened.

EXPLAIN

Why is fixation permanent here? Which line of *What this model assumes* makes it so — and what would have to change for A1 to reappear?

5 Drift is predictable on average

12 min

SET UP

$N = 50$, $p = 0.50$, Haploid; then the same in Diploid.

PREDICT FIRST

With M gene copies ($M = N$ haploid, $2N$ diploid), $H_t = H_0(1 - 1/M)^t$. Starting from $H_0 = 0.5$, calculate the heterozygosity left after 35 generations in each case.

RUN

Press **Run 10 Simulations** and read the *Genetic Diversity* chart: measured mean H (solid) against expected (dashed). Repeat in Diploid.

LOOK AT

the generation at which the measured curve has fallen to half its starting value.

EXPLAIN

The ten trajectories disagree about everything, yet the diversity curve follows the dashed line. Why is an average over loci so much more predictable than any one locus? What does that imply for how real studies measure genetic diversity?

6 What the starting frequency changes, and what it does not

10 min

SET UP

$N = 50$, Haploid. **Run 10 Simulations** at $p = 0.5$, then at $p = 0.1$.

PREDICT FIRST

Will heterozygosity fall at the same *proportional* rate in the two cases? Will A1 fix equally often?

RUN

Run both batches, twice each.

LOOK AT

the starting height of the diversity curve, how steeply it falls relative to that height, and how many runs fix A1.

EXPLAIN

Sort the quantities you looked at into those that depend on p and those that depend only on M.

7 Mutation against drift

12 min

SET UP

N = 50, Haploid, p = 0.50, **Mutation rate μ = 0.005**, and move G up to about 400.

PREDICT FIRST

Will the diversity curve still fall to zero? If not, where will it settle? (MSc: set $H_{t+1} = H_t$ in $\mathrm{E}[H_{t+1}] = (1 - 1/M)[(1 - 2\mu)^2 H_t + 2\mu(1 - \mu)]$ and solve.)

RUN

Press **Run 10 Simulations**. Then double N to 100 and run again.

LOOK AT

the level the diversity curve flattens at, and the summary line that compares it with the arithmetic.

EXPLAIN

Why does the curve flatten instead of falling to nothing? Why does mutation between just two alleles pull p towards 0.5? What does this tell you about the runs in steps 1–4, which all ended at 0 or 1?

DISCUSSION AND WORKSHEET QUESTIONS

1. BSc A population of 1,000 and a population of 20 both carry two neutral alleles at 50:50. Which is more likely to have lost one of them after 50 generations, and why?
2. BSc "Drift is just noise; selection is what really drives evolution." Use what you saw in this room to argue for or against.
3. BSc The room says evolutionary time is measured in generations, not years. Give an example of two species for which that distinction changes how fast drift acts.
4. MSc A population of 10,000 adults is refounded every spring by about 50 breeders. Which number belongs in M, and why does the room call its N an *effective* size?
5. MSc Explain why the expected heterozygosity curve is the same for every starting frequency, while the probability of fixing A1 is not.
6. MSc At N = 50 and μ = 0.005 the balance sits near H = 0.25. What happened when you doubled N? Relate your answer to the product $N\mu$.

CHECKING YOUR REASONING

Read this after you have written your own answers. These are what the model should show on most runs — not the only acceptable explanations.

- **Step 1.** The next generation is rarely exactly 0.50. Fifty draws at $p = 0.5$ have a standard deviation of $\sqrt{(0.25/50)} \approx 0.07$, so anything from about 0.36 to 0.64 is unremarkable. The change is sampling and nothing else.
- **Step 2.** At $N = 10$ a run usually reaches 0 or 1 within a few tens of generations; at $N = 200$ it wanders slowly and may still be unfinished after a few hundred. Which allele wins cannot be predicted: at $p = 0.5$ each is equally likely.
- **Step 3.** Expect about 5 of 10 to fix A_1 at $p = 0.5$ and about 2 of 10 at $p = 0.2$, with a batch-to-batch spread of one or two either way. Pooled over many batches the fraction approaches p.
- **Step 4.** Nothing happens: with $\mu = 0$ there is no way to remake A_1, so 0 and 1 are absorbing. A mutation, or a migrant (the Migration Room), would be needed to bring it back.
- **Step 5.** Haploid: $0.5 \times 0.98^{35} \approx 0.25$, so about half is gone after $0.69M \approx 35$ generations. Diploid: $0.5 \times 0.99^{35} \approx 0.35$. The measured mean tracks the dashed line; single loci do not.
- **Step 6.** Starting H differs (0.50 against 0.18), and A_1 fixes about half the time against about one time in ten. The proportional decay, $1/M$ per generation, is the same in both.
- **Step 7.** H settles near 0.25 instead of falling to 0; the equilibrium of the recursion is 0.247 at $M = 50$. At $N = 100$ it rises to about 0.33: more gene copies, slower loss, so the same mutation rate holds more variation.